jegyzet: SOGATs, by examples

1 Levels of abstraction
2 Langs w/o binders
3 Derivability in langs w/ binders
4. SO → FO
5 Admissibility in langs w/ binders

Coq $+, \times, \top, \bot, \Rightarrow$

Unit, Void, (), $\top$
$\_\times\_$ record, class $A\times(B\times C)$, $(A\times B)\times C$, ...
$\_+\_$ variant, subclass, union, $A,B \subseteq A+B$ enum $\top+\top+\top+\top$
Empty, $\bot$ $\bot\Rightarrow\bot$

$(a+b)^2 = a^2+2ab+b^2$

$A^B = B\Rightarrow A$
$3^2 = 9$

Def: ($A \cong B$ (A izomorf B-vel))
$f : \mathrm{Tm}\,A \to \mathrm{Tm}\,B$
$g : \mathrm{Tm}\,B \to \mathrm{Tm}\,A$
$f(g\,b) = b$
$g(f\,a) = a$

$\mathrm{Bool} \Rightarrow (A+B) \cong$
$(\mathrm{Bool}\Rightarrow A) + \mathrm{Bool}\times A\times B + (\mathrm{Bool}\Rightarrow B)$
true ↦ ... : A, false ↦ ... : A | true ↦ A, false ↦ B | A, B | true ↦ ... : B, false ↦ ... : B
$\cong A\times A + \mathrm{Bool}\times A\times B + B\times B$

Three := $\top + (\top+\top)$
$|\mathrm{Three}\times\mathrm{Bool}| = 6$
$|\mathrm{Three} + \mathrm{Bool}| = 5$
$|\mathrm{Three}\Rightarrow\mathrm{Bool}| = 2^3$
$|\mathrm{Bool}\Rightarrow\mathrm{Three}| = 3^2 = 9$

0 = false, 1 = true

	f1	f2	f3	...				
0	0	0	0	0	1	1	1	1
1	0	0	1	1	0	0	1	1
2	0	1	0	1	0	1	0	1

$(\mathrm{fst},\mathrm{snd}) : \mathrm{Tm}(A\times B) \cong \mathrm{Tm}\,A\times\mathrm{Tm}\,B : \_,\_$
$\mathrm{Tm}(A\Rightarrow B) \cong (\mathrm{Tm}\,A\to\mathrm{Tm}\,B) : \mathrm{lam}$
$\mathrm{Tm}(A+B) \cong \mathrm{Tm}\,A + \mathrm{Tm}\,B : \mathrm{inl},\mathrm{inr}$

$f : \mathrm{Tm}((A+B)\times C) \leftrightarrow \mathrm{Tm}(A\times C + C\times B) : g$
$f\,w = \mathrm{case}\,(\lambda a.\,\mathrm{inl}\,(a, \mathrm{snd}\,w))\,(\lambda b.\,\mathrm{inr}\,(\mathrm{snd}\,w, b))\,(\mathrm{fst}\,w)$
$g\,w = \mathrm{case}\,(\lambda ac.\,(\mathrm{inl}\,(\mathrm{fst}\,ac), \mathrm{snd}\,ac))\,(\lambda cb.\,(\mathrm{inr}\,(\mathrm{snd}\,cb), \mathrm{fst}\,cb))\,w$
$f(g\,w) \overset{?}{=} w$ → $w = \mathrm{inl}\,(a,c)$, → $w = \mathrm{inr}\,(c,b)$
$g(f\,w) \overset{?}{=} w$ → ..., → ...

$\mathrm{inl} : \mathrm{Tm}\,A \to \mathrm{Tm}(A+B)$
$\mathrm{inr} : \mathrm{Tm}\,B \to \mathrm{Tm}(A+B)$
$\mathrm{case} : (\mathrm{Tm}\,A\to\mathrm{Tm}\,C) \to (\mathrm{Tm}\,B\to\mathrm{Tm}\,C) \to \mathrm{Tm}(A+B)\to\mathrm{Tm}\,C$
$+\beta_1 : \mathrm{case}\,f\,g\,(\mathrm{inl}\,a) = f\,a$
$+\beta_2 : \mathrm{case}\,f\,g\,(\mathrm{inr}\,b) = g\,b$
~~$(\exists a.\,t = \mathrm{inl}\,a) \vee (\exists b.\,t = \mathrm{inr}\,b)$~~
$\boxed{+\eta\ (f : \mathrm{Tm}(A+B)\to\mathrm{Tm}\,C) \to f = \mathrm{case}\,(\lambda a.\,f(\mathrm{inl}\,a))\,(\lambda b.\,f(\mathrm{inr}\,b))}$

$\times\beta_1 : \mathrm{fst}\,(a,b) = a$
$\times\beta_2 : \mathrm{snd}\,(a,b) = b$
$\times\eta\ \ t = (\mathrm{fst}\,t, \mathrm{snd}\,t)$

$f(g(\mathrm{inl}\,(a,c))) \overset{+\beta_1\,\times\beta}{=}$
$f(\mathrm{inl}(\mathrm{fst}\,(a,c)), \mathrm{snd}\,(a,c)) =$
$f(\mathrm{inl}\,a, c) \overset{+\beta_1}{=}$
$\mathrm{inl}\,(a, \mathrm{snd}\,(\mathrm{inl}\,a, c)) \overset{\times\beta_1}{=}$
$\mathrm{inl}\,(a,c)$

$\boxed{(w : A+B) \to w = \mathrm{case}\,\mathrm{inl}\,\mathrm{inr}\,w}$
$(\lambda w.\,w) = \mathrm{case}\,\mathrm{inr}\,\mathrm{inr}$

$A+B \cong B+A$
$f = \mathrm{case}\,\mathrm{inr}\,\mathrm{inl}$
$g = \mathrm{case}\,\mathrm{inr}\,\mathrm{inl}$
$f(g\,w) \overset{?}{=} w = \mathrm{case}\,\mathrm{inl}\,\mathrm{inr}\,w$ ②
$= \mathrm{case}\,\mathrm{inr}\,\mathrm{inl}\,(\mathrm{case}\,\mathrm{inr}\,\mathrm{inl}\,w)$
$= \mathrm{case}\,\mathrm{inr}\,\mathrm{inl}\,(\mathrm{case}\,\mathrm{inr}\,\mathrm{inl}\,(\mathrm{case}\,\mathrm{inl}\,\mathrm{inr}\,w))$

$+\eta : (h : \mathrm{Tm}(A+B)\to\mathrm{Tm}\,C) \to h = \mathrm{case}\,(h\circ\mathrm{inl})\,(h\circ\mathrm{inr})$

$f\circ g : \mathrm{Tm}(A+B)\to\mathrm{Tm}(A+B)$
$\|{+\eta}$
$\mathrm{case}\,(\lambda a.\,f(g(\mathrm{inl}\,a)))\,(\lambda b.\,f(g(\mathrm{inr}\,b))) \overset{+\beta_1}{\underset{+\beta_2}{=}}$
$\mathrm{case}\,(\lambda a.\,f(\mathrm{inr}\,a))\,(\lambda b.\,f(\mathrm{inl}\,b)) \overset{+\beta_2}{\underset{+\beta_1}{=}}$
$\mathrm{case}\,(\lambda a.\,\mathrm{inl}\,a)\,(\lambda b.\,\mathrm{inr}\,b) =$
$\mathrm{case}\,\mathrm{inl}\,\mathrm{inr} = \lambda w.\,w$

$g(\mathrm{inl}\,a) =$
$\mathrm{case}\,\mathrm{inr}\,\mathrm{inl}\,(\mathrm{inl}\,a) =$
$\mathrm{inr}\,a$
$g(\mathrm{inr}\,b) =$
$\mathrm{case}\,\mathrm{inr}\,\mathrm{inl}\,(\mathrm{inr}\,b)$
$= \mathrm{inl}\,b$

Nullable $A = A + \top$

$\top + \top$

Boolη : $(f : \mathrm{Tm}\,\mathrm{Bool} \to \mathrm{Tm}\,C) \to f = \mathrm{ite}\,(f\,\mathrm{true})\,(f\,\mathrm{false})$

if x then true else false
$\|$
x

Fin. RING

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$A+(B+C) \cong (A+B)+C$
$\bot + A \cong A$
$A+B \cong B+A$
$A\times(B\times C) \cong (A\times B)\times C$
$\top\times A \cong A$
$A\times B \cong B\times A$
$(A+B)\times C \cong A\times C + B\times C$

$0, +, 1, *, ~~\times~~$
$B\times C\Rightarrow A \to B\Rightarrow C\Rightarrow A$
curry: $A^{B\times C} \cong (A^C)^B$: uncurry
$A^{B+C} \cong A^B \times A^C$
$A^\top \cong A$
$A^\bot \cong \top$
$\top^A \cong \top$

$\cong$ refl, sym, trans
$A \cong A$
$A \cong B \to B \cong A$
$A \cong B \to B \cong C \to A \cong C$
$\cong$ kongruencia, $+, \times, \Rightarrow$

$$\frac{A \cong A' \quad B \cong B'}{\begin{array}{c}A+B \cong A'+B' \\ A\times B \cong A'\times B' \\ A\Rightarrow B \cong A'\Rightarrow B'\end{array}}$$

$A+B \cong B+A$
↓
$(A+B)\times C \cong (B+A)\times C$

$\bigcirc\!\!\!\!-1$